SLEEP DEPRIVATION INCREASES TOXIN BUILDUP IN THE BRAIN

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ABSTRACT. In this paper, we develop and analyze ODE models of the glymphatic system, focusing on soluble and fibular amyloid-β, in order to investigate efficient strategies for obtaining sufficient sleep. Using our results, we examine whether it is possible to recover lost sleep and explore alternative sleep strategies for achieving adequate rest.

1. BACKGROUND/MOTIVATION

The exact functions and physiology of sleep remain an ongoing topic of investigation and discussion in medicine. However, recent findings have indicated that its purpose is, in part, to enable regular clearance of the metabolic waste products, or metabolites, which naturally accumulate in the interstitial space between brain cells. [XL13] One of the mechanisms that makes this clearance possible is known as the glymphatic system. This system is primarily active during sleep, specifically N3 or slow-wave sleep, when the interstitial space between brain cells expands. [RO20] This expansion and the ensuing glymphatic activity flood the brain with cerebrospinal fluid, which, by various pathways, carries metabolites into the blood and lymphatic system for processing. [TCJ15, PB18]

One metabolite of special interest is known as amyloid-β (Aβ), which is significant for its well-documented correlation with neurological disorders, including Alzheimer's disease (AD). [Gou15] In its soluble form, Aβ can be cleared from the interstitial fluid by the glymphatic system. However, by an aggregation process, Aβ monomers bind into toxic plaques characteristic of AD. The progression of this aggregation is not uniform, but rather evolves over the course of decades. Initially, the brain maintains a phase where Aβ remains largely soluble and available for glymphatic clearance. However, the emergence of fibrillar structures alters these dynamics. Established plaques effectively sequester soluble Aβ from the interstitial fluid and incorporate it into growing deposits. [?] This mechanism establishes a self-reinforcing cycle wherein the presence of plaques accelerates the capture of new monomers. Consequently, once significant fibrillation has occurred, the pathology becomes increasingly resistant to clearance, underscoring the importance of preventative measures during the early stages of accumulation. [TCJ15, Gou15]

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Empirical data suggests that Aβ dynamics are strongly connected to circadian rhythm. In healthy individuals, Aβ concentrations in the cerebrospinal fluid follow a distinct diurnal pattern, rising during wakefulness and falling during sleep. This oscillation appears to be driven by neuronal activity, which increases Aβ production during waking hours, balanced by glymphatic clearance during rest. Crucially, this healthy oscillation is not permanent. The amplitude of this rhythm dampens significantly with age and is nearly absent in individuals with fibrillar amyloid deposition. This empirical flattening of the Aβ curve suggests that as the brain ages, it loses its dynamic ability to reset Aβ levels overnight, leading to a chronic elevation in baseline concentration. The rate at which this occurs is thought to be dependent on many factors, including long-term sleep patterns. [Hua11]

An important question to consider is how lifestyle choices and other conditions impact the glymphatic system's efficacy in Aβ clearance. Although this is important to understanding AD pathology, empirical research on the matter is limited by the invasive nature of direct Aβ measurements from the cerebrospinal fluid and lack of reliable alternative means.[Hua12] Herein lies the aim of our model: to quantify the effects of lifestyle choices on glymphatic activity and resulting Aβ concentration in the brain. In particular, we explore the effects of differing sleep patterns on both soluble Aβ levels in the short term and fibular Aβ buildup in the long term.

1.1. Related Work

There are decades of precedent for a two-process sleep model, one that involves both circadian and homeostasis sleep regulation. [A22] Our model disregards circadian influences on sleep regulation in favor of isolating the effects of sleep patterns on homeostasis.

Inspired by the biological mechanism of the glymphatic system and by the nature of Aβ aggregation, we designed a compartmental model. [TCJ15, Irv08] However, it should be noted that our model simplifies the biological process by ignoring extracellular degradation clearance of Aβ and instead reducing this metabolite clearance to purely a result of glymphatic activity.

As aforementioned, the invasiveness of direct Aβ measurements from the cerebrospinal fluid limits human empirical data that might be used in generating and validating our model. However, some such data does exist and is supplemented by studies on rats and mice. [Hua11, ZB19]

2. Modeling

As mentioned above, we used a compartment model for soluble and fibular Aβ toxin levels in the brain, represented as A_s and A_f respectively.

Glymphatic activity primarily depends on space for fluid to flow in and clear metabolites. Our research indicated interstitial space in the brain during REM was nearly identical to wakefulness. [RO20] To simplify, we then decided not to model each sleep stage, instead only modeling deep sleep and non-deep sleep.

The brain enters deep sleep more during the earlier hours of sleep. As the night goes on, sleep becomes more dominated by REM. [Sun25] To preserve the cyclic nature of sleep, but also account for that decrease, we decided to use a decaying sine wave:

$$\sin(\theta t)e^{-\lambda t},$$

where θ is the frequency and λ is the exponential decay constant. Shifting and scaling this wave to make it realistic gives our final model for how the brain drifts in and out of deep sleep:

$$\frac{1}{2}(\sin(\theta t - 1.57) + 1)e^{-\lambda t}$$

We can then construct a piecewise function for glymphatic activity where activity is kept to a minimum during waking hours

$$G(x,t) = \begin{cases} \frac{1}{2}(\sin(\theta t - 1.57) + 1)e^{-\lambda t}A_s & \text{if asleep} \\ (.01)A_s & \text{if awake} \end{cases}$$

With our model of glymphatic activity, we then turned to modeling the production of Aβ in the brain. Since Aβ is a byproduct of brain cell metabolism, production is proportional to the level of brain cell activity.

Studies we found show that there is a natural 11% decrease during sleep [EAN02] in activity. In addition, metabolic activity can also be slowed by an over-abundance of waste in the brain. This means that there is an inverse relationship between current waste levels and created waste. We decided to model this using a carrying capacity k similar to predator-prey models. Thus, production becomes

$$P(x,t) = \frac{A_s}{k} \cdot \begin{cases} .89 & \text{if asleep} \\ 1 & \text{if awake} \end{cases}$$

The last piece of the equation is to determine how soluble waste converts into damaging fibular waste. This is composed of three terms. The first, cA_s^2, represents a process called primary nucleation. This is the rate at which soluble Aβ monomers form fibular with each other. Herein is a simplification of our model, as we categorize monomers as soluble and all polymers as non-soluble, when in fact Aβ aggregates exist on a spectrum of size and solubility. We justify this simplification with the slow rate of primary nucleation and the rapid toxification and secondary nucleation of Aβ polymers. The second mass action term, hA_sA_f represents this latter process of secondary nucleation whereby Aβ monomers bind to existing polymers. This is a much lower energy process and so it happens much more rapidly, though it is dependent on the concentration of fibular Aβ. The final term is a disaggregation term representing a concentration-dependent rate of disintegration of fibular Aβ polymers. [Irv08, Hua11, Gou15]

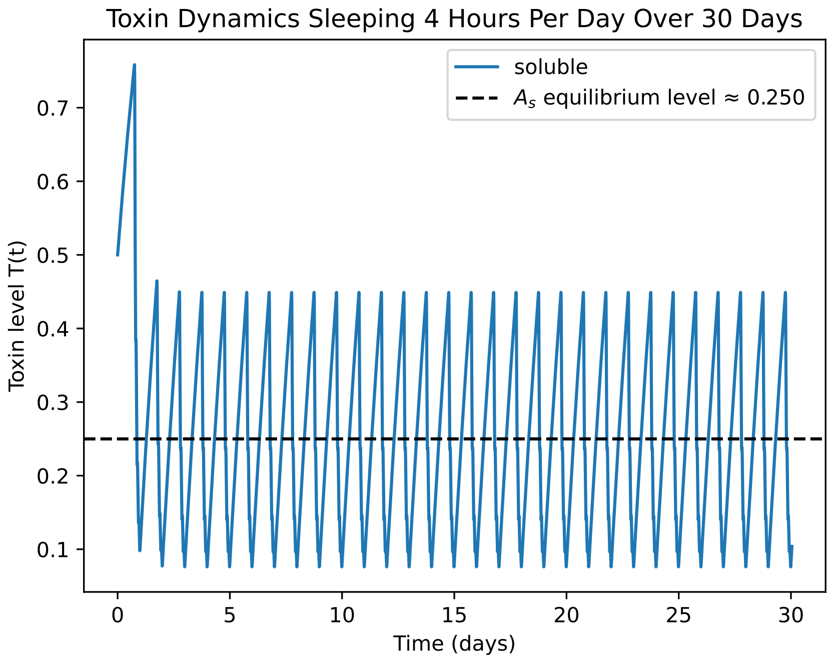

FIGURE 1. Levels of soluble toxins over the course of 30 days. The sawtooth behavior is expected as production of toxins during day-time hours increases levels and the Glymphatic systems clears them during sleep.

$$\text{agg}(x,t) = cA_s^2 + hA_sA_f - gA_f$$

Putting these pieces together gives the differential system where soluble toxin levels are increased by production ($P(x,t)$) and decreased by glymphatic clearance ($G(x,t)$) and the soluble toxins aggregating into fibular toxins ($agg(x,t)$). Fibular toxin levels are only increased by aggregation and never decrease [Hua11]. Specifically,

$$\frac{dA_s}{dt} = P(x,t) - G(x,t) - \text{agg}(x,t)$$
$$\frac{dA_f}{dt} = \text{agg}(x,t)$$

3. RESULTS

Running our model shows that solutions tend to settle within a certain bandwidth of Aβ levels. Solutions in this resting bandwidth oscillate according to that individuals sleep cycle, with Aβ levels reaching a maximum just before sleep and a minimum right after. All of this behavior is consistent with what we would expect based on the background research we did.

The specific resting bandwidth of the solutions is dependent on the amount of hours slept and not on the initial conditions. Changing the initial toxin level between 0.3, 0.5, and 0.7 all show that solutions settle on the same bandwidth (see Figure 2). This result suggests that there is an equilibrium state that oscillates around an average Aβ level. The average Aβ level remains consistent for each sleep pattern, again behaving as we expect based on our background research (see Figure 3).

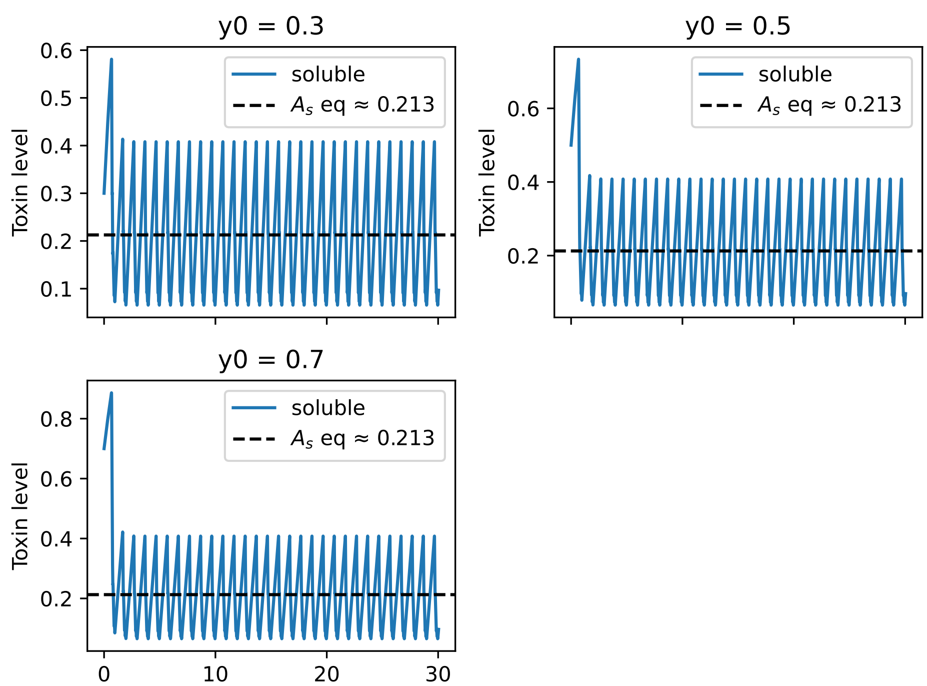

FIGURE 2. Over a 30-day period, all three initial conditions $y_0 = [0.3, 0.5, 0.7]$, converge to the same equilibrium under a regular 8-hour sleep schedule, indicating that the long-term toxin level is independent of the starting value.

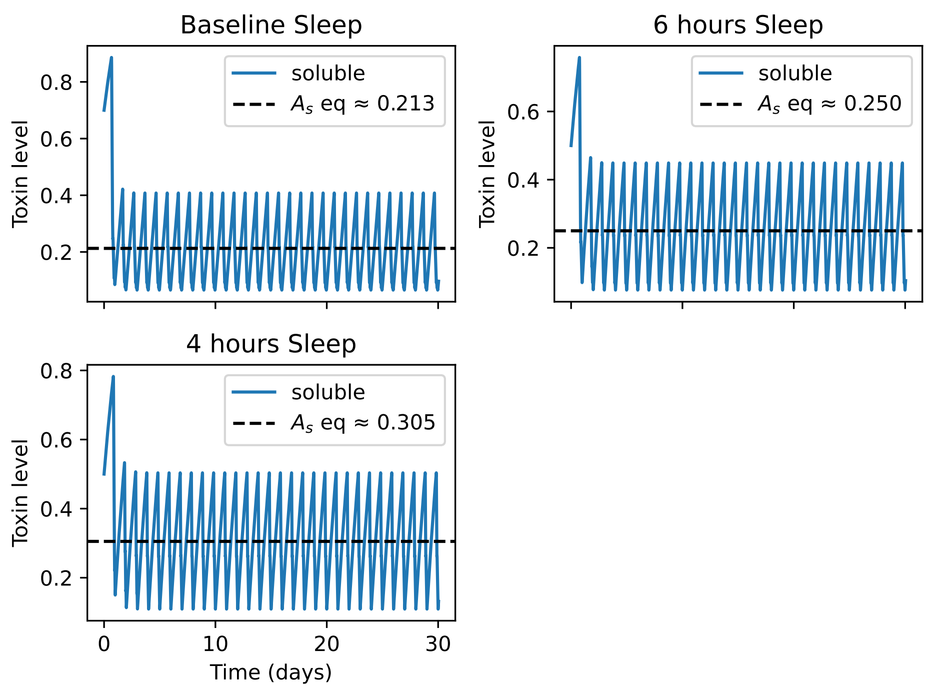

FIGURE 3. Over a 30-day period with the same initial condition, 4 hours of sleep leads to a higher toxin equilibrium than 6 hours, and 6 hours leads to a higher equilibrium than 8 hours.

Since we are interested in longer-term trends, we modeled individuals consistently getting 4, 6, and 8 hours of sleep. The model predicts that individuals who sleep less will have higher average Aβ levels than those who consistently get more sleep. The increase in Aβ levels increases exponentially as an individual sleeps less. This is also consistent with the fact that the Glymphatic system is more active during the earliest hours of sleep.

Because no benchmark model is available for comparison, we estimated the Lyapunov exponent using the SciPy *ivp_solver* as the stepper in order to find our system's stability. The results show that the Lyapunov exponent estimate is always negative and converges to zero over time. The Lyapunov exponent near zero means that the system is stable in the Lyapunov sense, meaning that if the initial state is close to the equilibrium, it will remain close to it.

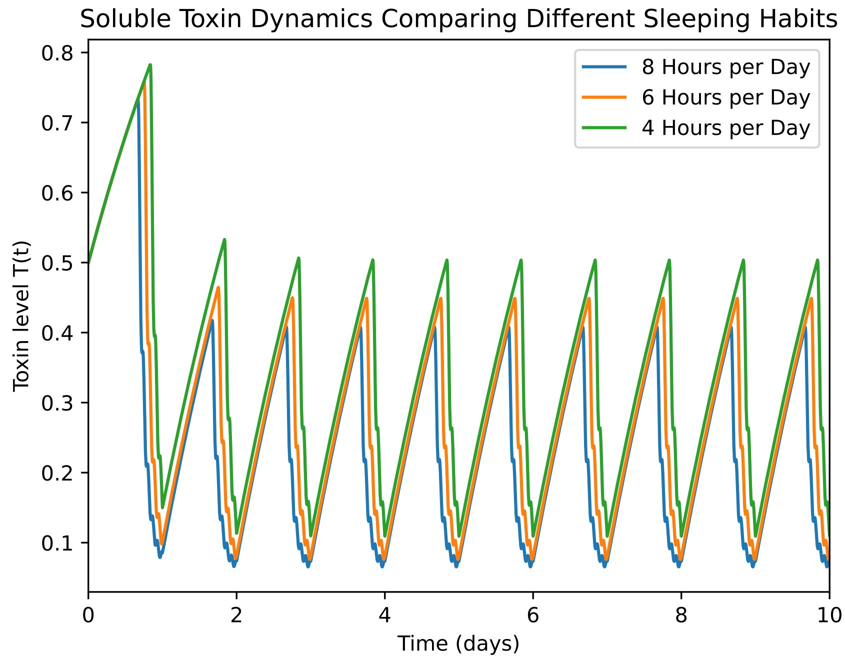

FIGURE 4. Over 30 days, getting 4 hours of sleep results in higher overall toxins levels than 8 hours of sleep. Even if an individual is given an initial condition with lower toxin levels, solutions will quickly move to a steady wave over the course of a couple of days.

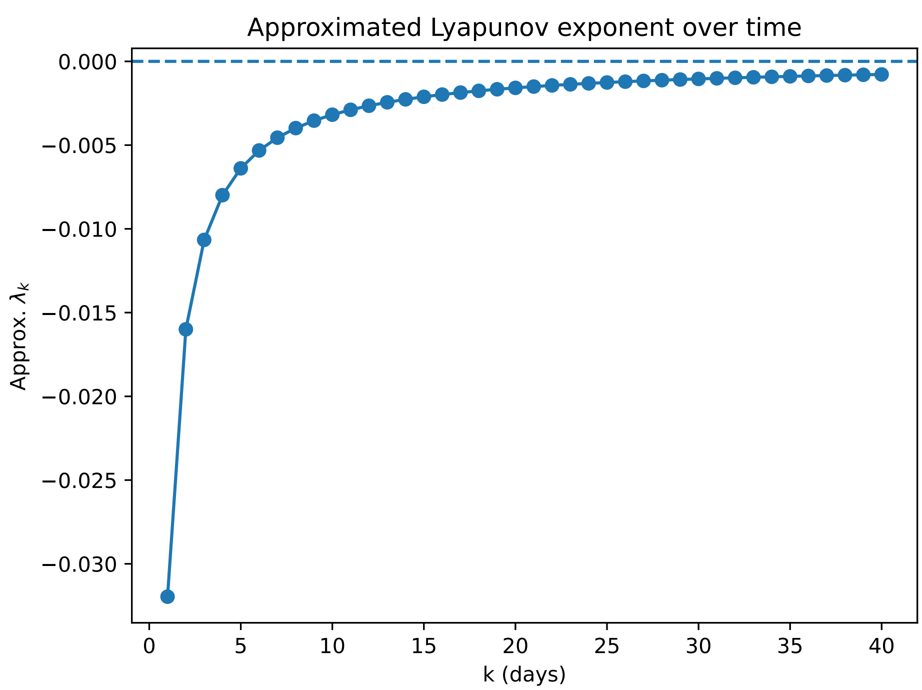

FIGURE 5. Lyapunov exponent approximation over time with $y_0 = 0.5$. The approximated Lyapunov exponents are all negative and converge to 0 over time.

4. Analysis/Conclusions

Overall, our compartment model successfully captures essential features of glymphatic dynamics and reproduces consistent patterns in Aβ clearance. The observed convergence to sleep-dependent equilibria, along with the rise in toxin levels under reduced sleep, supports the hypothesis that chronic sleep deprivation increases amyloid burden in the brain over time.

However, several simplifications limit our model's realism. The use of a discontinuous sleep-wake switching function introduces numerical artifacts at transition points, and the model does not distinguish between REM and non-REM sleep, despite the substantial differences between these stages. Future work should incorporate smoother sleep transitions, and a more accurate representation of nightly glymphatic behavior.

Despite these limitations, the model provides a valuable preliminary framework for quantifying the influence of sleep patterns on amyloid accumulation. It highlights sleep duration as a potentially important and modifiable factor in maintaining efficient glymphatic function.

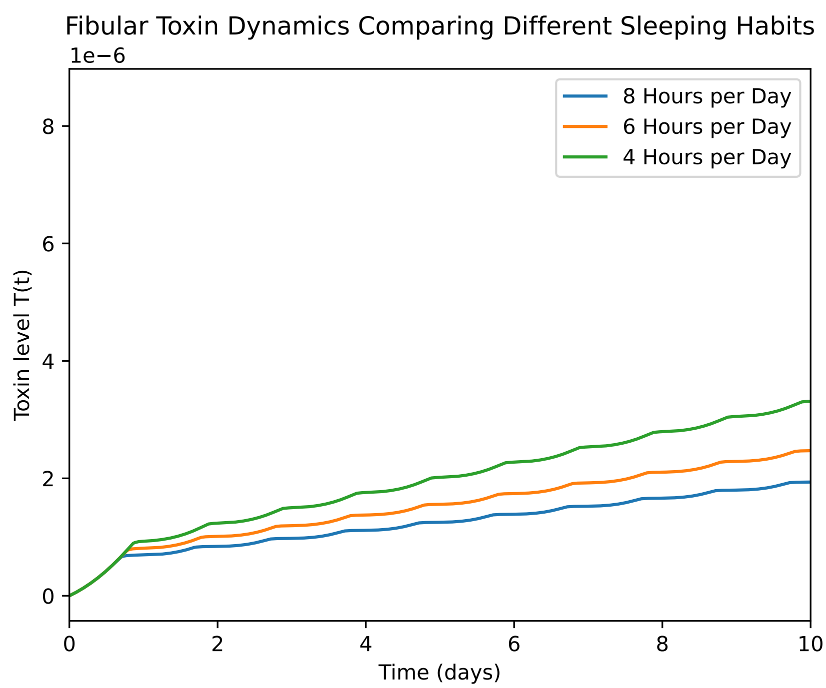

Figure 6. Fibular waste steadily increasing regardless of sleep. However, getting less sleep leads to fibular levels increasing at a faster rate.

Maintaining an equilibrium of toxin levels through the glymphatic system is essential to maintaining brain activity as the mind ages. Our results highlight that maintaining a consistent sleep schedule with at least 6 hours of sleep will contribute to keeping toxin levels stable and, by implication, help to mitigate the potential for neurological disorders.

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